CODE COVERAGE REPORT

Electronics Store Inventory Management System (EIMS) · Team 9

Test execution	Passed	Failed	Duration	Line coverage
Complete pytest suite	181	0	117.84 s	85%

1. Objective

This report records the line-coverage measurement for the EIMS application code using pytest-cov. Coverage indicates which executable statements were exercised by the test run; it does not, by itself, prove correctness or completeness of testing.

2. Test Environment and Command

Environment: Windows 11 (win32), Python 3.12.7, pytest 8.3.5, pytest-cov 7.1.0, virtual environment (.venv).

Command: pytest --cov=app --cov-report=term-missing --cov-report=html:htmlcov --cov-report=xml:coverage.xml

The run collected 181 tests. All 181 passed, with one dependency deprecation warning. Execution time was 117.84 seconds.

3. Overall Coverage Result

The coverage tool reported 1,139 statements, 172 missed statements, and 85% overall line coverage for the measured app package.

Source file	Statements	Missed	Coverage
app/__init__.py	0	0	100%
app/bootstrap.py	47	6	87%
app/config.py	6	0	100%
app/db.py	13	0	100%
app/main.py	495	91	82%
app/models.py	106	0	100%
app/monitoring/__init__.py	0	0	100%
app/monitoring/monitor.py	47	47	0%
app/monitoring/service.py	17	17	0%
app/repositories.py	154	4	97%
app/schemas.py	172	6	97%
app/security.py	71	1	99%
app/services.py	11	0	100%
TOTAL	**1,139**	**172**	**85%**

4. Coverage Observations

Highest coverage includes app/config.py, app/db.py, app/models.py and app/services.py (100% in this run). app/repositories.py and app/schemas.py each reported 97%, app/security.py reported 99%, app/bootstrap.py reported 87%, and app/main.py reported 82%.

The monitoring modules were not exercised: app/monitoring/monitor.py reported 0% (47 of 47 statements missed), and app/monitoring/service.py reported 0% (17 of 17 statements missed). These are the clearest areas to investigate for additional tests if monitoring is an in-scope feature.

The coverage output also identifies missed line ranges in several files. The detailed line-level view is available in the generated HTML report at htmlcov/index.html, and the machine-readable report is coverage.xml.

5. Limitations

This is statement/line coverage, not branch coverage. The command did not enable branch measurement. The 85% figure must therefore not be presented as branch coverage. Coverage is measured for the app package selected by --cov=app and reflects this test run and environment only.

6. Conclusion

The complete pytest suite passed all 181 tests and achieved 85% line coverage of the measured app package. The results show strong exercised coverage in several core modules, while the monitoring modules and some lines in app/main.py and app/bootstrap.py remain uncovered. Additional targeted tests can be considered where those paths are required.